

Predictive Zonotope Reduction: Sound and Bounded-Memory Uncertainty Monitoring for Robotics

Albert Author¹ and Bernard D. Researcher²

Abstract— Set-based methods like zonotopes give sound, formally-grounded over-approximations of state under uncertainty, which makes them attractive for safety-critical robotics. Online, however, they do not scale: each step adds generators, and the usual fix — reducing the zonotope to a fixed order — either blows up memory or, when done offline, forces pessimistic worst-case input assumptions. Recent work achieves bounded-memory online reduction but commits to a single reduction method for the entire run. We observe that the reduction applied now affects the tightness of the set many steps later, so the choice of method is a sequential decision rather than a one-time configuration. We frame online zonotope reduction as a control problem — the action at each step is which reduction method to apply — and solve it with Model Predictive Control. The framing is sound by construction: every reduction is a sound over-approximation, so any schedule is too, and the choice affects only tightness, never safety. To meet real-time monitoring budgets, we distill the policy into a neural ranking policy with DAGger that matches, and in our experiments exceeds, online MPC while running far faster. Built on RTLola, a mature monitor with verified semantics and a ROS adapter, our method produces tighter reachable sets and fewer false-positive interventions than any single-method baseline, with bounded memory and real-time inference

I. INTRODUCTION

Robots acting in the physical world must make decisions under uncertainty. A mobile robot moving through clutter or a manipulator arm moving near a human does not observe a single exact state, but a set of states consistent with noisy sensors, model error, and actuation uncertainty. Dealing with uncertainty requires methods computing possible reachable states, and if this set is too small, the robot may be allowed to perform an unsafe action. If it is too large, the safety monitor becomes overly conservative and triggers unnecessary stops or interventions. For safety-critical robotics, the central challenge is therefore not only to estimate uncertainty, but to track it soundly and tightly enough for online decisions. This also has impact on learning in robotics, as safety interruptions caused vacuously slow down and break training in realistic robot settings.

Zonotopes are a natural representation for this problem and are widely used from floating-point computation, neural network verification and computer graphics, and have a rich use for reachability analysis in cyber-physical systems [1], [2]. They provide compact, sound over-approximations of reachable state sets and are closed under the linear operations needed for uncertainty propagation.

However, their use in estimating uncertainty in cyber-physical systems suffers from a core trade-off between the performing these approximations online, and doing them offline. Doing it offline, as in tools such as CORA [1],

leads to *pessimistic* overapproximations as it has to consider all possible inputs which have to be presumed worst case, and not the actual ones that happen at runtime. On the other hand, approximating uncertainty with zonotopes online, with access to actually system inputs leads to a prohibitive memory blowup as every step adds generators, so the size of the zonotope representation grows without bound [3].

Recent work developing RLola [4] shows you can combine both of these approaches through online reduction with bounded memory by compressing zonotopes on the fly during online computations. But just like CORA [1] it commits to a single reduction method for the whole run. This misses a core insight: how you reduce the zonotope now, will impact how it will look in the future. Every reduction in zonotopes has impact both on the efficacy of reductions in the future, as well as what information is lost or preserved in the current step. Furthermore, the choice between all possible zonotope reduction methods is hard as there is a lot of them, and even more important: the best reduction is situation-dependent — different methods dominate in different regimes — so fixing one up front leaves tightness on the table, and worse, some methods quietly fail when their assumptions (e.g. invertibility) break.

In this paper, we show that we can improve uncertainty handling in monitoring by approaching zonotope reductions as a dynamic sequence problem. Unlike CORA [1] or RLola [4] approaches, which restricts the reduction method to be the same during the entire runtime of the system, we frame the problem as one of dynamic choice and allow different methods. To do it effectively, we frame this dynamic reductions as an optimal control problem, where the control action at each step is the selection of the optimal zonotope reduction algorithm. We show that a dynamic control theory based approach outperforms static zonotope reduction approaches. Our framing is safe by construction: every individual reduction is a sound over-approximation, so any sequence of choices preserves the safety conditions required for safety-critical environments.

To enable use in real-time systems we refine the method through several optimizations. We reduce the computational overhead of MPC through search methods through the MPC policy via beam search, and push this even more by distilling the control policy into a neural one. We show that by distilling the control policy built through MPC into a neural one we get both a better policy, and one which runs significantly quicker than MPC and which actually can be deployed for real-time systems.

We build on Kohn et al by using RTLola monitor that

provides us with strong formal guarantees. This gives us a strong foundation, as RTLola has seen extensive use in cyber-physical systems [5], [6], [7], [8], and allows us to provide formal guarantees about the memory bounds, semantics and soundness of zonotope operations [9], [10], [11]. We evaluate our method on the simulation of a 5-degrees-of-freedom robotic arm simulated in MuJoCo [12], with the sensor noise and uncertainty being modelled according to ISO-5725 [13].

We show that our approach outperforms others, both static zonotope reduction methods as seen in CORA and RLola. Our zonotopes are tighter, and the false positive rate is significantly lower. A lower FPR means that our method would cause less unnecessary emergency stops or task aborts - crucial both for safe operations in safety-critical environments as well as robot learning in uncertain environments where mistakes during exploration can cause damage or harm. Our method preserves soundness while reducing those false interventions. Furthermore, unlike probabilistic methods or Kalman Filter based approaches our method provides concrete formal guarantees about the over-approximations and makes it usable in safety critical environments. In comparison to offline methods, our method can take into account the actual inputs and give tighter expectations. This makes our work appropriate for safety-critical applications.

We give the following contributions

- We formulate bounded-memory online zonotope reduction as an optimal control problem in which the monitor's action at each step is the choice of a sound reduction operator. This captures the central observation that reduction is not merely a local compression step: the reducer chosen at time t changes the geometry of the zonotope available at later steps.
- We instantiate this formulation with a receding-horizon MPC policy that selects reducers by optimizing predicted future set tightness. To meet real-time monitoring budgets, we distill the control policy into a DAGger-trained neural ranking policy that selects among sound reducers at runtime.
- We implement predictive reducer selection in the RTLola monitoring framework with ROS integration and evaluate it on robotic systems under realistic measurement uncertainty. Compared with fixed-reducer baselines and offline/online variants, our method produces tighter reachable sets, reduces false-positive safety interventions, and achieves real-time inference while maintaining soundness and bounded memory.

II. BACKGROUND

A. Zonotopes

Math definition. Use in other domain

B. Runtime Verification

Runtime monitoring is an applied formal method that assures the safety of a running system by evaluating its behavior against a formal specification [14]. In particular, it has seen wide applications to cyber-physical systems,

especially in safety-critical settings [5], [6], [15], [16]. A particularly powerful paradigm of runtime verification is the stream-based approach [17], represented with tools such as TeSSLa [18], Striver [19] or RTLola [20]. Here, the process runs through a specification given in the form of terms of equations that relate input streams (that contain raw data such as sensor readings or actuator positions) to output streams (that transform and aggregate the incoming streams into information, or new streams themselves). The values on the output streams are then checked against trigger conditions that indicate faulty or dangerous situations and can be used to monitor or intervene into the operations of the system being observed.

a) RTLola: In this paper, we primarily focus on RTLola, which has seen extensive use in autonomous and safety-critical systems including UAVs [5], [6], [21], as well as deep integration into the robotics software stack via a ROS adapter [8]. The tool is validated in practice, supported on different hardware [22], [23], well documented [24], [25], and has an extensive formal methods foundations including memory bound guarantees, formally verified compilers and semantics [9], [10], [11]. We provide a detailed definition of RTLola syntax and semantics in Appendix TODO.

C. Model Predictive Control

Model Predictive Control (MPC) is the standard receding-horizon approach to optimal control. For a system $x_{k+1} = f(x_k, u_k)$, at each step k the controller observes the current state and solves a finite-horizon problem

$$\min_{u_k, \dots, u_{k+H-1}} \sum_{j=0}^{H-1} \ell(x_{k+j}, u_{k+j}) \quad (1)$$

where H is the horizon and ℓ the stage cost. Only the first input u_k^* is applied; the system advances, and the problem is re-solved from the new state at $k+1$. Solving over a horizon requires a model of how the system will evolve—in particular, a prediction of the future inputs over the next H steps.

In Section X we frame dynamic zonotope reduction as a MPC problem with the action being the choice of reduction method and the stage cost penalizing the size of the resulting set.

D. Learning from Experts

Imitation learning trains a policy to reproduce the decisions of an expert. In our setting the expert is the control policy developed through Model Predictive Control reduction policy and we distill it into a neural policy using DART?

III. RELATED WORK

Conformal prediction for runtime monitoring + Symbolic pruning

MPC for uncertainty - Tube MPC etc.

Runtime Verification and uncertainty. Lubeck people. Althoff people

the contributions should be offline + on-line

TEMPORAL LLM PLACEHOLDER

IV. UNCERTAINTY WITH ZONOTOPES

Problem setting - why uncertainty is everywhere and why its important to be handled. So you want to be safe, while still accounting for Uncertainty you have some approaches to measure in involve that uncertainty. This is also important for RL in robotics as learning in real-world environments needs to way to ensrue safety so you odnt crash the drone. Zonotopes for uncertainty in state as a way to do that.

Our definition of the uncertainty setup. We adapt the error model induced by the ISO5725 [13] standard by decomposing the measurement error into a constant, but unknown per-sensor off-set and a randomly varying per-measurement error. This decomposition directly correlates with the “trueness” and “precision” described in the ISO 5725.

A. Running Example

A small toy example for intuition - omni robot. Small omni robot as in Kohn et al. Give dynamics, environment. Figure with a picture + RTLola spec. Motivate the story a bit and explain the spec and what RTLola does here with zonotopes.

The RTLOLA spec is indendent as it is supposed to monitor the system - its not that hard to specify behaviour, and the tool is already mature and supported with ROS etc.

V. PREDICTIVE ZONOTOPE REDUCTIONS

Motivate the horizon and MPC as a good fit. Framing the Zonotope reduuctions as control decisions.

Framing of the problem as a control problem. Definition of the state, action, step etc. as if this was control theory.

We frame the online zonotope reduction as an online control problem. At each step we pick an action which is which method you pick.

The cost function is evaluated against the unrolled zonotopes with approximation (can be done bounded in memory). We then evaluate the trajectories against that uncompressed one.

Importantly - we preserve soundness! As any choice you select is locally sound, so is the full trace.

A. Offline Monitoring

rollout mpc with the oracle, find optimal thing

In offline cases we have the full trace of inputs - we can do exact MPC optimisation over that. Recording the trace doesnt remove uncertainty!

More horizon = more performance but at exponential cost
More methods = more flexibility, but most of them suck.
Shortly discuss all the CORA ones

B. Online Monitoring

In the full online setting, we dont know the inputs in the future. We have two ways to handle this. One is to defer the horizon in a way, store the +H zonotopes fully, while predictiong the -H zonotope as if it was the offline setting.

In case that doesnt work, we can predict the inputs as well. As we dont see input, so we have to predict them so we interpolate. We tryed out some ismple methods and they

TABLE I

INPUT-PREDICTION ABLATION FOR HORIZON-3, WIDTH-4 BEAM MPC. FORECAST NRMSE IS NORMALIZED AGAINST HOLD PREDICTION. ALL VARIANTS HAVE ZERO FALSE NEGATIVES AND ZERO EXECUTED FALLBACKS.

Predictor	nRMSE ↓	FPR (%) ↓	$\bar{L}$ ↓	ms/ev. ↓
Hold	1.000	0.568	1.31×10^{-6}	156.42
Linear	0.464	0.570	1.43×10^{-6}	158.85
Quadratic	0.488	0.565	1.28×10^{-6}	155.86

basically work quite decently, so we do the stupid simple one.

ablation study of known input vs different ways to predict inputs (interpolation methods) show pareto % of the known inputs

doing the full rollout too expensive, so we do beam search. Show pareto vs fully known

C. Monitoring via Distillation

While the MPC appraoch is better as we seen in Table X, it is also slow as it requires unrolling of the MPC and optimiziation. Runtime verification has to run at lightspeed level shit.

To preserve the flexibility, and improving the speed of inference we distill the control policy. The neural net is tiny as we optimize for speed, and the inputs are matrix information (so not even the input is needed). We distill the control policy into this and show that it works really good and really quick.

Doing a classifier doesnt work, as it is really unbalanced dataset, and picking the minority actions when they matter is really important. thats why we dont learn a classifier, but a ranker. this is a much richer signal and handles the unbalance better

Table comparing the MPC selection vs Neural slection. Showing that it is almost the same (or even better) while being much faster - and still sound (as the reducers are sound) and memory bounded.

VI. EVALUATION

We ask five questions. Does treating reduction as a sequential decision produce tighter sets and fewer false interventions than any fixed reducer (Sec. VI-B)? What does the predictive search cost, and how much of that cost is necessary (Sec. VI-C)? Does distillation preserve the accuracy of the MPC policy while meeting the monitor’s real-time budget (Sec. VI-D)? How do the policies behave on controlled, full-length fault traces (Sec. VI-E)? And which reducers do the controllers actually select (Sec. VI-F)? Section VI-G then reports a tail-remedy ablation that improves central behaviour without repairing the worst cases.

A. Experimental Setup

All experiments use the five-degree-of-freedom MuJoCo robot arm [12], the packaged RTLola safety monitor, and the ISO 5725 measurement-error decomposition [13].

We evaluate the seven binding transform bounds $b \in \{40, 80, 120, 150, 200, 250, 500\}$ against one exact unreduced reference per trace.

a) Metrics: Our primary loss $\bar{L}$ is the event-mean binding-native approximation loss, and FPR uses exactly negative reference events as its denominator. Because every reduction is individually sound, no method can produce a false negative by construction; accuracy differences are therefore differences in tightness and in the false interventions that looseness causes. Throughput is event-loop throughput in events per second. Traces advance at $\Delta t = 0.1$ s, so a monitor must sustain 10 Hz to keep pace with the system it observes; we treat that rate as the real-time budget throughout. Relative throughput statements use paired cells, because concurrent execution affects absolute timing.

b) Methods: The canonical comparison covers the fixed Girard, Scott, PCA, and Combastel reducers; recorded-future terminal beam MPC (MPC-B); the one-event full-width teacher (MPC-F); causal predictive linear MPC (MPC-L); and the Geometry15 policy trained on 148 clean teacher trajectories (G15/Clean148). MPC-B and MPC-F consume information a deployed monitor does not have and are accuracy references, not deployable policies.

c) Cohorts: Generalization evidence comes from 20 held-out nominal random-waypoint traces (seeds 100–119) of 500 events each, giving 140 seed–budget cells per method. The four full 2,340-event figure-eight traces — nominal, drift, geofence, and drift plus geofence — are controlled descriptive case studies and do not estimate a fault population. The guard confirmation of Sec. VI-G uses 20 previously untouched nominal seeds, 328–347, and the same four fixed cases. We froze the selection rule and all eligibility thresholds before observing those cells.

B. Dynamic Selection Against Fixed Reducers

Table II and Fig. 1 answer the first question, and the separation is not marginal. Pooled over all 140 nominal cells, every fixed reducer produces a false positive on at least a third of negative events — Girard 33.3 %, Combastel 34.9 %, Scott 41.4 %, PCA 63.0 % — while every dynamic selector stays at or below 2.58 %. The tightness gap is larger still: the best fixed reducer has a median $\bar{L}$ of 6.6×10^{-4} against 1.6×10^{-7} for MPC-L, roughly three and a half orders of magnitude. Scott is the clearest illustration that a single method cannot be trusted across regimes: it is competitive in the literature, yet here its interval fallback fails every budget-40 cell and its median loss reaches 5.2×10^7 .

Figure 1 places this in the plane that matters for deployment. The fixed reducers occupy the fast, loose region; the MPC variants occupy the tight, slow region and fall inside the shaded band where the monitor cannot keep up with a 10 Hz system. The distilled policy is the only method in the lower-right quadrant, and Sec. VI-D quantifies that claim.

C. The Cost of Predictive Search

Two design choices set the price of the MPC policy: how far the rollout looks ahead, and how wide the beam

TABLE II
HELD-OUT NOMINAL GENERALIZATION AT $b = 150$

Method	$\bar{L} \downarrow$	FPR (%) $\downarrow$	Ev./s $\uparrow$	RT
<i>Fixed reducer</i>				
Girard	7.5×10^{-4}	33.08	132.3	✓
Scott	7.9×10^7	40.83	99.9	✓
PCA	2.8×10^{-1}	65.33	117.9	✓
Combastel	6.8×10^{-4}	29.78	194.4	✓
<i>Offline oracle</i>				
MPC-B	1.7×10^{-7}	1.48	2.1	×
MPC-F	2.1×10^{-7}	1.55	8.5	×
<i>Online</i>				
MPC-L	1.5×10^{-7}	1.44	2.3	×
G15/Clean148	1.9×10^{-7}	1.48	96.7	✓

$\bar{L}$ and throughput are medians over 20 seeds; FPR is the macro mean with a paired-seed bootstrap. Bold marks the best value in each column among the causal methods; no method holds more than one. The offline oracles consume recorded future inputs or an exhaustive teacher and are accuracy references only. Real time is throughput ≥ 10 Hz, the trace event rate. Fig. 2 gives the full budget sweep.

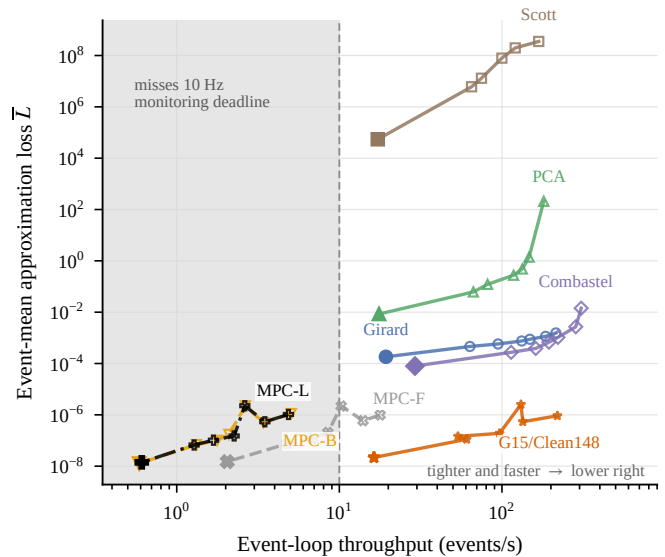


Fig. 1. Accuracy against cost on the held-out nominal seeds. Each trail connects one method’s seven transform bounds, from $b = 40$ (open marker, highest throughput) to $b = 500$ (filled marker); both axes are medians over 20 seeds. The shaded band marks throughputs below the 10 Hz trace event rate, where the monitor cannot keep pace with the system. Every MPC variant lies inside that band at every bound; the distilled G15/Clean148 policy attains MPC-scale loss outside it.

is. Table III(a) separates them on nominal seeds 60–64 at $b = 150$, run single-worker so the timings are contention-free.

The horizon is worthless without a beam. Every $W=1$ configuration returns exactly the same loss, 2.91×10^{-7} , and the same FPR regardless of H : with one surviving candidate the rollout cannot change which reducer wins now, only how long it takes to confirm it. Widening the beam to $W=2$ captures most of the available gain, cutting loss to 1.27×10^{-7} to 1.70×10^{-7} , and further widening yields no consistent improvement — at $H=4$ the loss rises again from

TABLE III
COST OF THE PREDICTIVE SEARCH

(a) Horizon and beam width								
H	Median $\bar{L}$ ($\times 10^{-7}$)				Median events/s			
	W=1	W=2	W=4	W=8	W=1	W=2	W=4	W=8
1	2.91	1.67	1.59	1.59	45.9	31.8	20.6	20.5
2	2.91	1.27	1.40	1.51	31.8	20.2	11.5	8.3
4	2.91	1.48	1.61	1.68	19.1	11.3	6.2	3.7
8	2.91	1.70	1.28	1.63	10.4	6.0	3.2	1.7

(b) Stage objective			
Objective	Median $\bar{L}$	Ratio	Tighter
Terminal (deployed)	4.2×10^{-7}	1.000	12
Cumulative (matched)	2.7×10^{-7}	0.867	16

(a) Nominal seeds 60–64 at $b = 150$, 80 cells, single worker so timings are contention-free; values are medians over five seeds. Bold marks the tightest and the fastest configuration in the grid, which are never the same one. $W=1$ is column-constant because a width-one beam admits no choice for the horizon to inform.

(b) Terminal against matched cumulative stage objective over the 28 paired fixed cells.

$W=2$ to $W=8$. The cost, by contrast, grows monotonically and steeply: throughput falls from 45.9 s^{-1} at $(H, W) = (1, 1)$ to 1.7 s^{-1} at $(8, 8)$, a factor of 26. Only the seven cheapest configurations meet the 10 Hz budget, and the configuration used elsewhere in this paper, $(4, 4)$, is not among them. This is the concrete reason distillation is necessary rather than merely convenient.

Table III(b) compares the terminal stage objective we deploy with a matched cumulative objective on the 28 paired fixed cells. The cumulative objective is tighter on 16 of 28 cells with a median loss ratio of 0.867. We read this as no meaningful separation at the scale that distinguishes dynamic from fixed selection, and we retain the terminal objective because it is what the distillation teacher was defined against.

D. Distillation and the Real-Time Budget

The distilled policy is the result the paper turns on, and the held-out cells support it on both axes. Its median paired loss is $1.258\times$ that of MPC-L — within a factor of two of a policy that sees a full beam rollout — while MPC-L retains only 2.62 % of its paired throughput. The real-time claim is stronger than a median: across all 140 nominal cells, MPC-L never exceeds 7.2 s^{-1} and MPC-B never exceeds 6.8 s^{-1} , so neither meets the 10 Hz budget in a single cell, whereas G15/Clean148 never falls below 12.3 s^{-1} and meets it in every cell. Its median throughput is 89.4 s^{-1} .

Figure 2 shows that this holds across the sweep rather than at one bound. In panel (a) the distilled policy tracks MPC-L within the width of the line at every budget; in panel (c) it tracks the *fixed* reducers instead, sitting an order of magnitude above the oracle band. Panel (b) exposes the one place the distillation is imperfect: at $b = 80$ the learned policy reaches 7.25 % macro FPR against 1.90 % for MPC-L, and its worst individual cell reaches 41.1 %. The learned

policy is therefore tight on average and occasionally over-triggers, which is the failure direction we prefer but not one we can claim to have eliminated.

E. Controlled Figure-Eight Case Studies

Tables IV and V give the exact values for the four full-length figure-eight cases. These are single fixed traces, so they are a controlled record of behaviour under specific faults, not a population estimate; we read them for consistency with the nominal result rather than as independent evidence. They are consistent: the fixed reducers remain three to eight orders of magnitude looser at every bound, and Scott is unavailable in all 56 of its cells. The dynamic policies hold FPR at or below 1.1 % on every fault variant except the two learned-policy excursions at $b = 120$ and $b = 500$, which are the same over-triggering behaviour visible in Fig. 2(b).

F. Reducer-Selection Preferences

Table VI reports which ordinary reducers the controllers select. All of them choose Scott on 85 % to 89 % of valid reduction decisions, and the learned policies choose Girard on roughly 9 % to 11 %. Scott’s dominance here is worth stating plainly alongside Table II, where static Scott is the worst method in the comparison: a reducer that is catastrophic when applied unconditionally is the right choice most of the time when the alternative is available for the steps where it is not. That is the sequential effect the paper argues for, though these are selection frequencies and not a causal decomposition — a reducer’s utility depends on where it sits in the sequence. The learned policies almost never choose PCA, and Vote3-Guarded shifts 1.6 percentage points from Girard to Combastel relative to Vote3.

G. Three Learned Policies: Distillation, Voting, and a Symbolic Guard

The distilled ranking policy of Sec. VI-D is the simplest way to make the teacher run in real time, and it is not the only one. We evaluate two further learned policies on a cohort of 140 nominal cells, seeds 328–347, that no policy in this paper was trained or tuned on. Vote3 takes a plurality vote among three independently optimized 5 %-DAgger specialists. Vote3-Guarded adds a symbolic override on top of that vote: when the winning margin is at most one, it compares the current-event native loss of the reducers that received a vote and takes the best, breaking the deadlock with a measurement rather than a learned preference. The three policies therefore form a progression — distillation, ensembling, ensembling with a symbolic tiebreak — against the same MPC-L teacher.

Table VII places them. Vote3-Guarded is the tightest of the three across the bulk of the distribution, at $1.19\times$ MPC-L in the median cell and $2.51\times$ at the ninetieth percentile, and it roughly halves the deployed policy’s macro false-positive rate, from 3.21 % to 1.86 %, while sustaining 69.0 s^{-1} . The margin over the distilled policy in the bulk is small — 2 % at both quantiles — but the direction is consistent, and the margin over unguarded Vote3 is not small: the guard moves

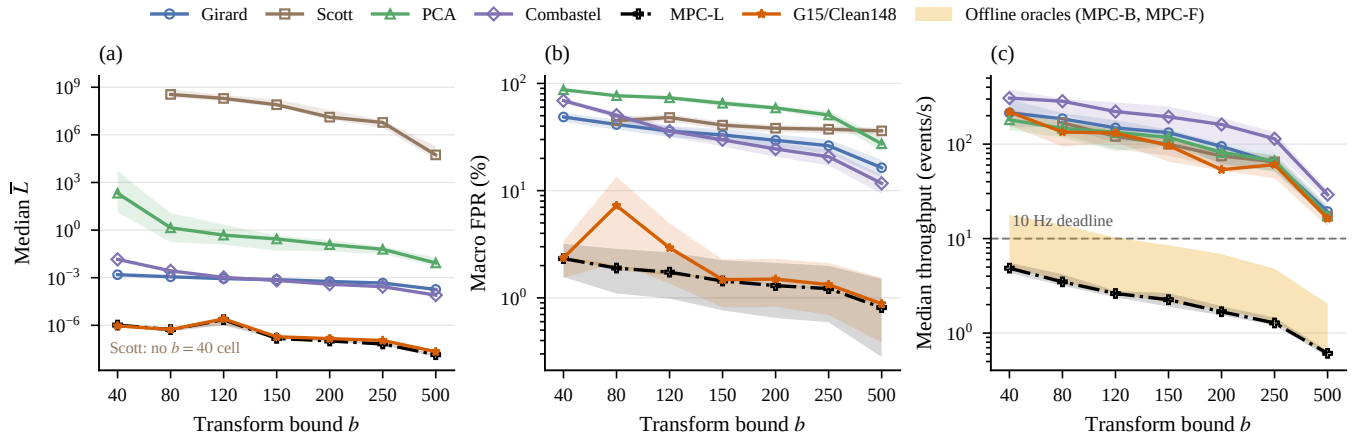


Fig. 2. Held-out nominal generalization over seeds 100–119, 20 traces per method and budget. (a) Median event-mean loss; (b) macro FPR; (c) median event-loop throughput. Shaded bands carry different meanings by panel: in (a) and (c) they are the interquartile range across the 20 seeds, a measure of variability; in (b) it is the paired-seed bootstrap 95% interval, a measure of uncertainty in the estimate. The tan band spans the two offline oracles MPC-B and MPC-F, which are collapsed because they are indistinguishable from MPC-L in (a) and (b). All vertical axes are logarithmic; the dashed line in (c) is the 10 Hz trace event rate. Scott has no budget-40 cell because its interval fallback fails there.

TABLE IV
APPROXIMATION LOSS $\bar{L}$ ON THE CONTROLLED FIGURE-EIGHT TRACES

Trace	b	Fixed			Offline oracle		Online			
		Girard	PCA	Combast.	MPC-B	MPC-F	MPC-L	G15	Vote3	V3-Guard
Figure-8	40	1.1×10^{-3}	2.3×10^2	4.3×10^{-1}	4.7×10^{-7}	5.6×10^{-7}	5.9×10^{-7}	5.6×10^{-7}	5.7×10^{-7}	5.6×10^{-7}
	80	9.8×10^{-4}	2.2×10^2	7.7×10^{-2}	7.1×10^{-7}	3.5×10^{-7}	1.8×10^{-6}	3.3×10^{-7}	3.1×10^{-7}	6.7×10^{-7}
	120	9.2×10^{-4}	6.5×10^1	2.8×10^{-2}	4.0×10^{-6}	4.6×10^{-6}	4.7×10^{-6}	4.1×10^{-4}	3.3×10^{-6}	3.3×10^{-6}
	150	8.9×10^{-4}	1.9×10^1	1.6×10^{-2}	1.2×10^{-6}	1.6×10^{-7}	1.1×10^{-6}	1.5×10^{-7}	1.5×10^{-7}	1.5×10^{-7}
	200	8.2×10^{-4}	2.7×10^{-1}	9.1×10^{-3}	1.3×10^{-7}	7.7×10^{-8}	6.4×10^{-7}	9.3×10^{-8}	9.3×10^{-8}	9.1×10^{-8}
	250	7.7×10^{-4}	1.5×10^1	5.4×10^{-3}	7.7×10^{-7}	7.6×10^{-8}	2.9×10^{-7}	5.4×10^{-8}	1.5×10^{-7}	6.8×10^{-8}
	500	6.0×10^{-4}	3.7	1.3×10^{-3}	1.7×10^{-8}	7.7×10^{-9}	2.5×10^{-8}	3.1×10^{-4}	3.3×10^{-5}	8.0×10^{-6}
Figure-8 drift	40	9.1×10^{-4}	4.8×10^2	4.0×10^{-1}	3.6×10^{-7}	3.1×10^{-7}	1.2×10^{-6}	4.8×10^{-7}	4.9×10^{-7}	4.8×10^{-7}
	80	8.2×10^{-4}	1.1×10^2	6.7×10^{-2}	2.5×10^{-7}	5.2×10^{-7}	8.1×10^{-7}	2.8×10^{-7}	4.2×10^{-7}	2.3×10^{-7}
	120	7.6×10^{-4}	6.5	2.5×10^{-2}	3.6×10^{-6}	6.0×10^{-6}	4.1×10^{-6}	3.6×10^{-4}	3.3×10^{-6}	3.3×10^{-6}
	150	7.2×10^{-4}	3.7	1.6×10^{-2}	1.8×10^{-7}	1.2×10^{-7}	6.3×10^{-7}	1.2×10^{-7}	1.3×10^{-7}	1.2×10^{-7}
	200	6.9×10^{-4}	3.4	8.4×10^{-3}	2.3×10^{-7}	6.3×10^{-8}	1.2×10^{-7}	7.8×10^{-8}	7.8×10^{-8}	7.5×10^{-8}
	250	6.3×10^{-4}	4.5×10^{-1}	5.5×10^{-3}	5.1×10^{-8}	2.0×10^{-7}	5.6×10^{-7}	4.5×10^{-8}	7.0×10^{-8}	4.6×10^{-8}
	500	5.1×10^{-4}	2.5	1.3×10^{-3}	3.6×10^{-8}	1.1×10^{-8}	2.5×10^{-8}	3.2×10^{-4}	1.1×10^{-4}	5.6×10^{-6}
Figure-8 geofence	40	1.2×10^{-3}	7.1×10^2	4.3×10^{-1}	7.2×10^{-7}	1.0×10^{-6}	7.0×10^{-7}	6.1×10^{-7}	6.2×10^{-7}	6.1×10^{-7}
	80	1.1×10^{-3}	4.9×10^2	7.4×10^{-2}	5.2×10^{-7}	6.6×10^{-7}	5.4×10^{-7}	3.5×10^{-7}	3.0×10^{-7}	6.9×10^{-7}
	120	1.0×10^{-3}	4.1×10^2	2.7×10^{-2}	5.2×10^{-6}	4.9×10^{-6}	3.9×10^{-6}	5.3×10^{-4}	3.3×10^{-6}	3.3×10^{-6}
	150	9.8×10^{-4}	2.2×10^1	1.7×10^{-2}	8.4×10^{-7}	2.9×10^{-7}	4.3×10^{-7}	1.6×10^{-7}	1.6×10^{-7}	1.6×10^{-7}
	200	9.2×10^{-4}	1.9×10^{-1}	9.3×10^{-3}	2.5×10^{-7}	8.2×10^{-8}	3.7×10^{-7}	9.8×10^{-8}	9.8×10^{-8}	9.4×10^{-8}
	250	8.6×10^{-4}	2.8×10^1	5.6×10^{-3}	6.7×10^{-7}	8.2×10^{-8}	2.6×10^{-7}	5.6×10^{-8}	5.7×10^{-8}	5.6×10^{-8}
	500	6.7×10^{-4}	6.0	1.4×10^{-3}	1.8×10^{-8}	9.6×10^{-9}	3.7×10^{-8}	3.4×10^{-4}	3.0×10^{-5}	6.4×10^{-6}
Figure-8 drift + geofence	40	1.0×10^{-3}	4.6×10^2	4.4×10^{-1}	5.1×10^{-7}	5.1×10^{-7}	9.8×10^{-7}	5.3×10^{-7}	5.4×10^{-7}	5.3×10^{-7}
	80	9.2×10^{-4}	1.3×10^2	7.4×10^{-2}	3.4×10^{-7}	1.5×10^{-6}	4.2×10^{-7}	3.0×10^{-7}	5.2×10^{-7}	2.4×10^{-7}
	120	8.7×10^{-4}	6.8	2.8×10^{-2}	4.9×10^{-6}	6.0×10^{-6}	5.5×10^{-6}	4.7×10^{-4}	3.3×10^{-6}	3.3×10^{-6}
	150	8.2×10^{-4}	5.1	1.7×10^{-2}	7.9×10^{-7}	2.5×10^{-7}	8.9×10^{-7}	1.3×10^{-7}	1.4×10^{-7}	1.3×10^{-7}
	200	7.6×10^{-4}	3.2	8.9×10^{-3}	2.3×10^{-7}	2.9×10^{-7}	1.3×10^{-7}	8.2×10^{-8}	8.2×10^{-8}	7.9×10^{-8}
	250	7.2×10^{-4}	2.8	5.5×10^{-3}	4.8×10^{-8}	2.0×10^{-7}	2.6×10^{-7}	4.7×10^{-8}	5.0×10^{-8}	4.7×10^{-8}
	500	5.6×10^{-4}	2.2	1.3×10^{-3}	4.8×10^{-8}	1.7×10^{-8}	4.8×10^{-8}	3.5×10^{-4}	7.8×10^{-5}	7.6×10^{-6}

Lower is better; bold marks the best value within each column block of a row, and a block whose entries all tie is left unmarked. MPC-B and MPC-F consume recorded future inputs or an exhaustive teacher and are accuracy references rather than deployable policies, so they are ranked among themselves rather than against the causal policies. Scott is omitted: its interval fallback fails in all 56 of its cells. Table V reports the corresponding false-positive rates.

the median cell from $1.48\times$ to $1.19\times$ and the ninetieth percentile from $4.18\times$ to $2.51\times$. Ensembling alone costs accuracy in the centre; ensembling with the guard recovers

it. Against G15/Clean148 cell by cell (Fig. 3(c)), Vote3-Guarded is tighter in 91 of 140 cells with a median paired ratio of 0.948, reduces mean FPR by 1.35 pp with a bootstrap

TABLE V
FALSE-POSITIVE RATE (%) ON THE CONTROLLED FIGURE-EIGHT TRACES

Trace	b	Fixed			Offline oracle		Online			
		Girard	PCA	Combast.	MPC-B	MPC-F	MPC-L	G15	Vote3	V3-Guard
Figure-8	40	51.67	97.48	94.27	0.00	0.00	0.00	0.00	0.00	0.00
	80	41.79	95.90	90.51	0.00	0.00	0.00	0.00	0.00	0.00
	120	33.46	93.80	86.75	0.00	0.00	0.00	10.30	0.00	0.00
	150	26.88	93.72	83.08	0.00	0.00	0.00	0.00	0.00	0.00
	200	23.46	76.97	77.14	0.00	0.00	0.00	0.00	0.00	0.00
	250	16.37	91.37	73.33	0.00	0.00	0.00	0.00	0.00	0.00
	500	2.22	83.42	44.96	0.00	0.00	0.00	7.39	0.00	0.00
Figure-8 drift	40	12.32	97.05	92.94	0.00	0.00	0.00	0.00	0.00	0.00
	80	5.32	95.37	88.36	0.00	0.00	0.00	0.00	0.00	0.00
	120	1.37	93.10	83.04	0.00	0.00	0.00	0.63	0.00	0.00
	150	1.79	91.36	79.99	0.00	0.00	0.00	0.00	0.00	0.00
	200	0.00	89.31	72.78	0.00	0.00	0.00	0.00	0.00	0.00
	250	0.00	76.51	65.98	0.00	0.00	0.00	0.00	0.00	0.00
	500	0.00	76.09	29.91	0.00	0.00	0.00	0.00	0.00	0.00
Figure-8 geofence	40	57.96	96.52	92.10	0.71	0.88	0.83	0.65	0.65	0.65
	80	45.34	94.34	86.91	0.71	0.71	0.24	0.41	0.35	0.47
	120	37.32	91.45	81.72	0.24	0.47	0.47	18.40	0.35	0.29
	150	32.96	91.33	76.65	0.47	0.24	0.35	0.29	0.29	0.24
	200	28.07	67.39	68.46	0.18	0.18	0.12	0.41	0.41	0.35
	250	24.47	88.09	63.21	0.12	0.24	0.06	0.18	0.12	0.18
	500	21.23	77.12	24.59	0.12	0.12	0.12	13.74	5.48	1.83
Figure-8 drift + geofence	40	25.73	96.64	91.96	0.84	0.90	0.84	1.02	1.02	1.02
	80	21.18	94.72	86.74	0.84	0.42	0.84	0.72	0.84	0.72
	120	20.52	92.14	80.68	0.84	0.84	0.48	16.20	0.48	0.66
	150	20.52	90.16	77.20	0.66	0.60	0.60	0.36	0.42	0.36
	200	20.16	87.82	68.99	0.78	0.60	0.66	0.42	0.42	0.42
	250	19.38	73.25	61.25	0.48	0.36	0.48	0.36	0.30	0.42
	500	18.36	72.77	24.36	0.36	0.48	0.48	11.28	6.78	0.42

Lower is better; columns and the bold convention match Table IV. Most rows leave the online block unmarked because all four policies reach 0.00 % there; bold appears only where they separate.

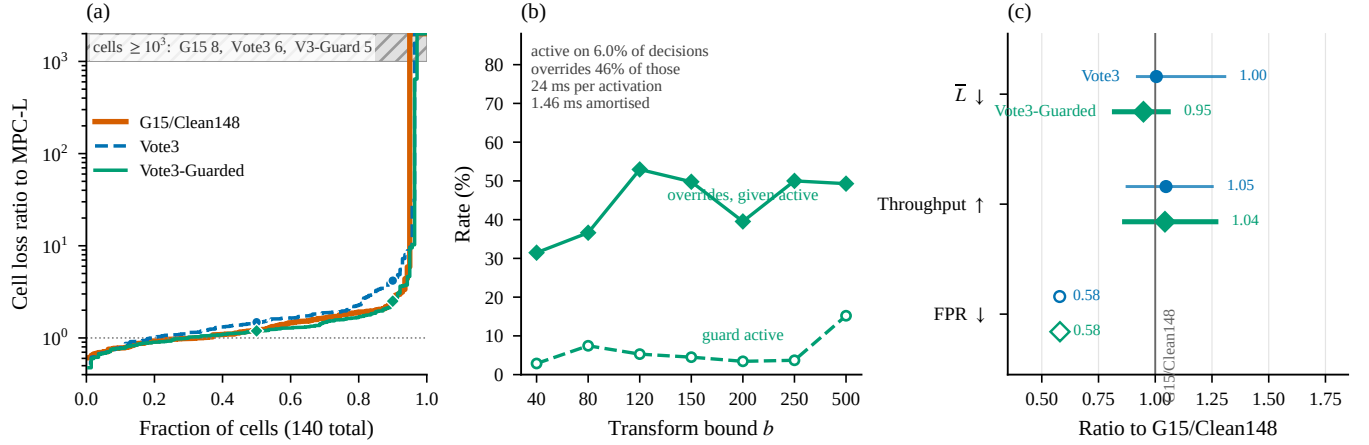


Fig. 3. The learned policies on the 140 untouched nominal confirmation cells, seeds 328–347. (a) Quantile function of the per-cell loss ratio to MPC-L: each curve reads left to right as the fraction of cells at or below a given ratio, markers sit at the median and ninetieth percentile, and the hatched band collects the cells beyond the preregistered severe threshold, whose counts are given inside it. (b) Activity of the symbolic guard against transform bound: the fraction of over-bound decisions on which it engages, and the fraction of those engagements in which it overrides the vote. (c) Both voting policies relative to the deployed G15/Clean148, with the good direction marked on each row. Loss and throughput are per-cell paired ratios shown as median and interquartile range over the 140 cells; FPR has zero-valued cells and is therefore the ratio of macro means, drawn as an open marker with no dispersion bar.

interval of $[-2.69, -0.25]$ that excludes zero, and retains $1.04 \times$ its throughput.

The override is sparse and effectively free (Fig. 3(b)). It activates on 6.03 % of over-bound nominal decisions, rising

TABLE VI
POOLED REDUCER-SELECTION FREQUENCIES (%)

Controller	Girard	Scott	PCA	Combastel
<i>Canonical nominal seeds 100–119</i>				
MPC-B	6.24	85.71	2.87	5.18
MPC-L	6.21	85.91	2.88	5.01
MPC-F	7.89	84.83	2.08	5.20
G15/Clean148	10.46	88.86	0.00	0.68
<i>Vote3 confirmation seeds 328–347</i>				
MPC-L	6.23	85.28	3.65	4.84
G15/Clean148	10.19	88.76	0.14	0.91
Vote3	10.58	87.67	0.00	1.75
Vote3-Guarded	8.99	88.91	0.00	2.09

Percentages condition on valid ordinary reduction decisions and exclude under-bound `none`, fallback, and infeasible decisions, so rows sum to 100; bold marks each controller’s most-selected reducer. Each row aggregates 69,100–69,120 decisions. The two panels use distinct nominal seed cohorts and are interpreted within, not across, panels.

TABLE VII
LEARNED POLICIES ON THE UNTOUCHED CONFIRMATION COHORT

Policy	Loss ratio to MPC-L						
	$\bar{L}$	p50	p90	p95	Sev.	FPR (%)	Ev./s
MPC-L	1.9×10^{-7}	1.00	1.00	1.00	0	1.41	2.1
G15/Clean148	2.2×10^{-7}	1.22	2.54	6.24×10^6	8	3.21	71.0
Vote3	3.7×10^{-7}	1.48	4.18	10.75	6	1.86	75.6
Vote3-Guarded	2.8×10^{-7}	1.19	2.51	9.53	5	1.86	69.0

140 cells, seeds 328–347, at seven transform bounds. $\bar{L}$ and throughput are medians; the quantile columns are per-cell paired ratios to the MPC-L teacher; Sev. counts cells beyond the preregistered 10^3 severe threshold; FPR is the macro mean. Bold marks the best value among the three real-time policies; MPC-L is the accuracy reference and does not meet the 10 Hz budget. G15/Clean148 distills the teacher into one ranking policy, Vote3 votes among three DAGger specialists, and Vote3-Guarded adds the symbolic override to that vote.

TABLE VIII
PREREGISTERED PROMOTION GATES

Preregistered gate	G15 (<i>deployed</i>)	Vote3	Vote3-Guarded
No severe cell	×	8 ×	6 ×
Budget medians $\leq 1.25 \times$	–	– ×	1.37 ✓
Throughput $\geq 0.5 \times$	–	– ✓	1.05 ✓
Mean FPR ≤ 0.5 pp	–	– ✓	–1.35 ✓
Worst-trace FPR ≤ 5 pp	–	– ×	5.06 ×
No cell failure	✓	0 ✓	0 ✓
All cells available	✓	140 ✓	140 ✓
No interval fallback	✓	0 ✓	0 ✓
Deployed under frozen rule	✓ (fallback)	×	×

Scored on the untouched nominal confirmation cohort, with the realized value beside each verdict. Thresholds and the selection rule were frozen before these cells were observed. Four gates are absolute properties of a policy and are also reported for the deployed G15/Clean148; the remaining four are defined as comparisons against it and are dashed in its column. FPR entries are percentage-point differences against G15/Clean148, where negative is an improvement.

to 15.2% at $b = 500$ where the specialists disagree most often, and changes the vote in 45.6% of those activations. Each activation costs 24.2 ms, but amortized over all decisions that is 1.46 ms, which is why the guarded policy is

no slower than the unguarded one. A comparison the reader should read as an ablation rather than a headline: against unguarded Vote3 the guard improves 115 of 140 cells and ties two, at a median paired ratio of 0.885.

The severe tail is the one place where no learned policy is clean, and it is worth stating precisely which policy owns it. Figure 3(a) plots the quantile function of the per-cell loss ratio to MPC-L; all three curves sit within a small factor of the teacher until roughly the ninety-fifth percentile, then diverge sharply. Vote3-Guarded has the fewest cells beyond the preregistered severe threshold of 10^3 , five, against six for Vote3 and eight for the deployed G15/Clean148 — so the guard reduces the severe count relative to both, and the deployed policy has the worst tail of the three. The effect is concentrated: seed 342 is severe for all three policies, and the guard’s severe cells span three seeds against the deployed policy’s five. Where the guard does regress it does so at the loosest bound, creating two new severe cells at $b = 500$ with absolute losses 1.044 and 2.538 at seeds 328 and 332 against 1.69×10^{-8} and 1.77×10^{-8} for Vote3. This is consistent with the hypothesis that a locally favourable override can steer a later reduction adversely — the same sequential coupling that motivates the method — but the experiment does not isolate a causal mechanism. On the four controlled figure-eight cases the guard is unambiguously better than the vote it corrects: zero severe cells against three, worst loss ratio to MPC-L falling from 4.38×10^3 to 322, and pooled FPR from 0.566% to 0.254%.

What the preregistered rule decided: Table VIII scores the challengers against the eight gates frozen before the confirmation cells were observed, with the deployed policy scored on the four gates that are absolute properties rather than comparisons against itself. Vote3-Guarded clears six of eight and is the only policy to clear the budget-median gate that Vote3 fails. It fails two. The first is the severe-tail gate — which, as Table VII shows, the deployed fallback fails more severely; it was written as an eligibility condition on challengers, not as a property the incumbent had been shown to satisfy. The second is the worst-trace FPR gate: one cell, seed 328 at $b = 500$, regresses by 9.94 pp against the 5 pp limit. That gate records only the largest regression, and the distribution it summarizes is strongly asymmetric in the favourable direction — the same policy’s three largest improvements are -40.0 , -39.9 and -39.6 pp.

We do not relax either gate after the fact, and under the frozen rule G15/Clean148 remains the deployed policy. We state the tension plainly rather than resolve it by revision: on this cohort Vote3-Guarded is the better policy on every axis we measure except worst-case, including the tail statistic the failed gate was written to protect, and promoting it is a matter for a fresh preregistration on a new cohort rather than for a reinterpretation of this one. MPC-L remains the accuracy reference with zero severe cells and the lowest macro FPR, at 3.14% of G15/Clean148’s paired nominal throughput.

VII. CONCLUSION

VIII. LIMITATIONS

The ML setup could be more interesting nad involved - especially direct zonotope data. We require the user to write a specification that monitors the system.

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